

Bill of Materials		MCP22301 Sink Board	
Part Number: EV71V36A Assy Number: 02-01257 PCB Number: 04-12321 Assy Revision: 1 Creation Date: 1/29/2025		 MICROCHIP	
Designator	Description	Manufacturer	Part Number
C1	CAP CER 1uF 100V 20% X7T SMD 0603	Murata	GRM188D72A105ME01J
C3	CAP CER 100nF 50V 10% X7R SMD 0603 AEC-Q200	AVX Corporation	06035C104K4T2A
C4	CAP CER 2.2uF 100V 10% X7R SMD 0805	TDK Corporation	C2012X7R2A225K125AC
C5, C6, C16	CAP CER 10uF 16V 20% X5R SMD 0805	TDK Corporation	C2012X5R1C106M085AC
C7, C8	CAP CER 1uF 50V 10% X7R SMD 0805	Samsung Electro-Mechanics	CL21B105KBFNNNE
C9, C11, C18, C19, C20, C21, C24	CAP CER 0.1uF 16V 10% X7R SMD 0603	Taiyo Yuden	EMK107B7104KA-T
C10, C14, C15	CAP CER 1uF 50V 10% X5R SMD 0603	Taiyo Yuden	UMK107B7105KA-T
C12, C13	CAP CER 220pF 50V 5% NP0 SMD 0603	KEMET	C0603C221J5GACTU
C17, C22	CAP CER 2.2uF 16V 80% Y5V SMD 0603	Yageo	CC0603ZRY5V7BB225
C23	CAP CER 10pF 50V 5% NP0 SMD 0603	KEMET	C0603C100J5GACTU
D1	DIO ZENER MMSZ5241BQ-7-F 11V 500mW SMD SOD-123 AEC-Q101	Diodes Incorporated	MMSZ5241BQ-7-F
D2	DIO RECT 1N4448 855mV 500mA 75V SOD-323	Diodes Incorporated	1N4448WS-7-F
D3	DIODE SCKY PMEG6010ER 530mV 1A 60V SMD SOD-123	NXP Semiconductors	PMEG6010ER,115
D4	DIO RECTARR BAV99 1V 215mA 70V SOT-23-3	Comchip Technology	BAV99-G
D5, D7, D8, D9, D10, D11, D12, D13	DIO LED GREEN 2.1V 30mA 6mcd Clear SMD 0603	Lite-On	LTST-C190GKT
D6	DIO LED RED 1.8V 20mA 7.1mcd Diffuse SMD 0603	OSRAM Opto Semiconductors	LS L29K-G1J2-1-Z
FB1, FB2	FERRITE 500mA 220R SMD 0603	Murata	BLM18AG221SN1D
J1	CON USB3.1 TYPE-C FEMALE TH R/A	Würth Electronics	632723300011
J2, J3	CON HDR-2.54 Male 1x2 Gold 5.84MH TH VERT	FCI	68000-202HLF
J4	CON TERMINAL 5.08mm 1X2 Female 16-30AWG 13.5A TH RA	TE Connectivity	282836-2
J5	CON HDR-2.54 Male 1x6 Gold 5.84MH TH R/A	FCI	68016-106HLF
J6	CON HDR-2.54 Male 1x8 Gold 6.00MH TH R/A solder directly	MULTICOMP PRO	MC34753
L1	INDUCTOR 15uH 650mA 20% WE-MAPI SERIES SMD 1212	Würth Elektronik	74438335150
Q1, Q2	TRANS FET P-CH SiR5607DP 60V 90.9A 104W PPAK SO-8	Vishay	SIR5607DP-T1-RE3
Q3, Q4	TRANS FET N-CH 2N7002-7-F 60V 170mA 370mW SOT-23-3	Diodes Inc	BSN20-7, 2N7002-7-F
R1, R2	RES TKF 300R 1% 1/4W SMD 1206	ROHM	MCR18EZH3000
R3	RES TKF 100k 1% 1/10W SMD 0603 AEC-Q200	Panasonic Electronic Components	ERJ-3EKF1003V
R4, R13	RES TF 10k 1% 1/16W SMD 0603	TE Connectivity Passive Product	CPF0603F10KC1
R5	RES TKF 39k 1% 1/16W SMD 0402 AEC-Q200	D9 Vishay Dale	CRCW040239K0FKED
R6, R7, R8, R11, R25, R37, R38, R39	RES TF 10k 1% 1/10W SMD 0402 AEC-Q200	Vishay	MCS0402MC1002FE000
R9	RES TKF 330R 1% 1/10W SMD 0603	Panasonic	ERJ-3EKF3300V
R12	RES TKF 226k 1% 1/10W SMD 0603 AEC-Q200	Panasonic Electronic Components	ERJ-3EKF2263V
R14, R27	RES TKF 200k 1% 1/10W SMD 0402 AEC-Q200	Panasonic Electronic Components	ERJ-2RKF2003X
R15, R18, R23, R40	RES TKF 1k 1% 1/10W SMD 0402	Panasonic	ERJ-2RKF1001X
R19, R20, R21, R22	RES TKF 4.7k 1% 1/10W SMD 0402 AEC-Q200	Panasonic Electronic Components	ERJ-2RKF4701X
R24, R32, R35	RES TKF 470R 1% 1/16W SMD 0402	Walsin Technology Corporation	WR04X4700FTL
R26, R33	RES TKF 100k 1% 1/10W SMD 0402 AEC-Q200	Panasonic	ERJ-2RKF1003X
R28	RES TKF 300k 1% 1/10W SMD 0402	Panasonic	ERJ-2RKF3003X
R29	RES TKF 390K 1% 1/16W SMD 0402	Yageo	RC0402FR-07390KL
R30	RES TF 510k 1% 1/10W SMD 0402 AEC-Q200	Panasonic Electronic Components	ERJ-2RKF5103X
R31	RES MF 0.005R 1% WW SMD 2512	Bourns Inc.	CRE2512-FZ-R005E-3
R34	RES TKF 681k 1% 1/16W SMD 0402 AEC-Q200	Vishay	CRCW0402681KFKED
R36	RES TKF 20k 1% 1/10W SMD 0402 AEC-Q200	Panasonic Electronic Components	ERJ-2RKF2002X
SW1	SWITCH ROTARY 8POS 24V 0.5A RM100002BCB TH	C&K Components	RM100002BCB
Microchip Parts			
U1	MCHP ANALOG SWITCHER Buck 2 to 24V MCP16331T-E/CH SOT-23-6	Microchip Technology	MCP16331T-E/CH
U2	MCHP ANALOG LDO 3.3V MCP1703AT-5002E/CB SOT-23A-3	Microchip	MCP1703AT-3302E/CB
U3	MCHP INTERFACE USB PD CONTROLLER MCP22301-2I/KYX VQFN40	Microchip Technology	MCP22301-2I/KYX
U4	MCHP ANALOG CURRENT SENSE AMP MCP6C02T-100E/CHY SOT-23-6	Microchip Technology	MCP6C02T-100E/CHY
Mechanical Parts			
JP1	MECH HW JUMPER 2.54mm 1x2	Sullins Connector Solutions	SPC02SYAN
LABEL1	LABEL, ASSY W/REV LEVEL (SMALL MODULES) PER MTS-0002	Raynen	10010276
PAD1, PAD2, PAD3, PAD4	MECH HW RUBBER PAD Hemisphere D6.4 H2.1 Black	Bumper Specialties Inc.	BS25BL07X30RP
Do Not Populate Parts (DNP)			
C2	CAP CER 0.1uF 16V 10% X7R SMD 0603	Taiyo Yuden	EMK107B7104KA-T
R10, R16	RES TKF 0R <0.01R 1/5W SMD 0402	Vishay	CRCW0402000020EDHP